

Quiz (Carolina Reaper)

Q1. What is the standard form of a quadratic function?

- (A) $ax^2 + bx + c$
 (B) $ax^2 - bx + c$
 (C) $ax^2 + bx - c$
 (D) $ax^2 - bx - c$
 (E) $ax + bx + c$

Q2. A shipping company's cost to transport x items is given by $C(x) = 2x^2 + 5x + 10$. What is the value of a ?

- (A) 2
 (B) 5
 (C) 10
 (D) 15
 (E) 20

Q3. The time it takes to travel from City A to City B is given by $t(x) = x^2 - 4x + 5$, where x is the speed. What is the minimum time?

- (A) 1 hour
 (B) 2 hours
 (C) 3 hours
 (D) 4 hours
 (E) 5 hours

Q4. The quadratic function $f(x) = ax^2 + bx + c$ has a vertex at $(2, 3)$. What is the value of b ?

- (A) $-4a$
 (B) $-2a$
 (C) 0
 (D) $2a$
 (E) $4a$

Q5. A traffic flow model is given by $f(x) = 3x^2 - 2x + 1$, where x is the number of cars. What is the maximum flow?

- (A) 1 car
 (B) 2 cars
 (C) 3 cars
 (D) 4 cars
 (E) 5 cars

Q6. The quadratic function $g(x) = x^2 + bx + c$ has a y -intercept at $(0, 2)$. What is the value of c ?

- (A) 1
 (B) 2
 (C) 3
 (D) 4
 (E) 5

Q7. A delivery company's revenue is given by $R(x) = 2x^2 + 3x - 10$, where x is the number of packages. What is the break-even point?

- (A) 2 packages
 (B) 3 packages
 (C) 4 packages
 (D) 5 packages
 (E) 6 packages

Q8. The quadratic function $h(x) = ax^2 + bx + c$ has a vertex at $(-2, 1)$. What is the equation of the axis of symmetry?

- (A) $x = -1$
 (B) $x = -2$
 (C) $x = -3$
 (D) $x = -4$
 (E) $x = -5$

Open Ended Questions (FRQ)

Q9. A transportation company's cost to transport x items is given by $C(x) = ax^2 + bx + c$. If the cost to transport 2 items is 200 and the cost to transport 4 items is 400, find the values of a , b , and c . Assume $a \neq 0$.

Q10. A traffic flow model is given by $f(x) = 2x^2 - 3x + 1$, where x is the number of cars. Find the maximum flow and the number of cars at which it occurs. Show all work and explain your reasoning.

Chef's Tips

- Tip 1: Make sure to check for vertex form when solving quadratic equations.
- Tip 2: Consider using graphing calculators to visualize the quadratic functions.
- Tip 3: Encourage students to use real-world applications to illustrate the concepts.